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संरक्षण — आईटी सुरक्षा अनुरूपता
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अपेक्षाएँ
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(पहला पुनरीक्षण)

Information Security, Cybersecurity
and Privacy Protection —
Requirements for the Competence
of IT Security Conformance
Assessment Body Personnel
Part 1 Overview and Concepts
(First Revision)

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NATIONAL FOREWORD

This Indian Standard (Part 1) (First Revision) which is identical to ISO/IEC 19896-1 : 2025 'Information security, cybersecurity and privacy protection — Requirements for the competence of IT security conformance assessment body personnel — Part 1: Overview and concepts' issued by the International Organization Standardization (ISO) and International Electrotechnical Commission (IEC) was adopted by the Bureau of Indian Standards (BIS) on the recommendations of the Information Systems Security and Privacy Sectional Committee and approval of the Electronics and Information Technology Division Council.

This Indian Standard is published in several parts. The other parts in this series are:

Part 2 Knowledge, skills and effectiveness requirements for ISO/IEC 19790 testers

Part 3 Knowledge, skills and effectiveness requirements for ISO/IEC 15408 evaluators

This standard was first published in 2019 and was identical with ISO/IEC 19896-1 : 2018. This revision has been brought out to align it with the latest version of ISO/IEC 19896-1 : 2025.

The main changes are as follows:

- a) the document has been restructured
 - subclauses related to experience, education and effectiveness have been deleted;
- b) technical changes have been introduced
 - competence concepts for the validators and the reviewers have been added;
- c) deleted elements of competence, experience, education and effectiveness except for knowledge and skills according to comments from ISO/CASCO.

The text of ISO/IEC standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions and terminologies are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard', and
- b) Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

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Introduction

The objective of the ISO/IEC 19896 series is to provide the fundamental concepts related to the topic of the competence of personnel responsible for performing information technology (IT) product security evaluations and conformance testing as well as those personnel performing review and validations. The ISO/IEC 19896 series provides the framework and the specialized requirements that specify the minimum competence of personnel performing IT product security evaluations and conformance testing, reviews and validation using International Standards. This document provides fundamental information to users of ISO/IEC 19896-2 and ISO/IEC 19896-3.

In pursuit of this objective, the ISO/IEC 19896 series comprises the following:

- a) the terms and definitions relating to the topic of competence in IT product security testers and evaluators;
- b) the fundamental concepts relating to competence in IT product security and conformance testing and evaluation;
- c) the minimum competence requirements for IT product security testers and evaluator to conduct IT product testing/evaluation;
- d) the terms and definitions relating to the topic of competence in IT product security validators and reviewers;
- e) the fundamental concepts relating to competence in IT product security validation and reviews; and
- f) the minimum competence requirements for IT product security validators and reviewers to conduct IT product validation/review.

The ISO/IEC 19896 series is of interest to:

- g) information security conformance-testing and evaluation specialists;
- h) information security review bodies for evaluation;
- i) information security validation authorities for conformance-testing;
- j) information security conformance-testing and evaluation laboratories;
- k) vendors or technology providers whose IT products can be the subject to information security conformance-testing or evaluations; and
- l) organizations offering professional credentials or recognitions.

The ISO/IEC 19896 series is organized into parts to address the competence of testing and evaluation personnel as follows.

This document provides an overview of the definitions, fundamental concepts and a general description of the framework used to communicate the competence concepts for certain specialized areas. This document aims to provide the fundamental knowledge necessary to use the framework presented in the other parts of the ISO/IEC 19896 series.

ISO/IEC 19896-2 describes the minimum set of competence requirements at each competency level for conformance testers and validators working with ISO/IEC 19790 and ISO/IEC 24759.

ISO/IEC 19896-3 describes the minimum set of competence requirements at each competency level for information security evaluators and reviewers working with the ISO/IEC 15408 series and ISO/IEC 18045.

*Indian Standard***INFORMATION SECURITY, CYBERSECURITY AND PRIVACY
PROTECTION — REQUIREMENTS FOR THE COMPETENCE OF IT
SECURITY CONFORMANCE ASSESSMENT BODY PERSONNEL****PART 1 OVERVIEW AND CONCEPTS***(First Revision)***1 Scope**

This document establishes an organized set of concepts and relationships to understand the competency requirements for information security conformance-testing and evaluation specialists, thereby establishing a basis for shared understanding of the concepts and principles central to the ISO/IEC 19896 series across its user communities.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org/>

3.1**competence**

ability to apply knowledge and skills to achieve intended results

[SOURCE: ISO/IEC 17024:2012, 3.6]

3.2**conformance-tester****tester**

individual assigned to perform test activities in accordance with a given conformance testing standard and associated testing methodology

Note 1 to entry: An example of such a standard is ISO/IEC 19790 and the testing methodology specified in ISO/IEC 24759.

3.3
evaluation laboratory
testing laboratory
laboratory

organization with a management system providing evaluation or testing work in accordance with a defined set of policies and procedures and utilizing a defined methodology for testing or evaluating the security functionality of IT products

Note 1 to entry: These organizations are often given alternative names by various evaluation authorities. For example, IT Security Evaluation Facility (ITSEF), Common Criteria Testing Laboratory (CCTL), Commercial Evaluation Facility (CLEF).

[SOURCE: ISO/IEC TS 23532-1:2021, 3.1, modified — testing laboratory and laboratory have been added as preferred terms; definition has been modified to cover testing; Note 2 to entry has been removed.]

3.4
evaluation method

set of one or more evaluation activities for application in a specific context

[SOURCE: ISO/IEC 15408-1:2022, 3.42]

3.5
evaluation methodology

structured framework or system that organizes and applies various *evaluation methods* ([3.4](#))

3.6
evaluator

individual assigned to perform evaluations in accordance with a given evaluation standard and associated *evaluation methodology* ([3.5](#))

Note 1 to entry: An example of evaluation standards is the ISO/IEC 15408 series. An example of an associated evaluation methodology is given in ISO/IEC 18045.

3.7
knowledge

facts, information, truths, principles or understanding acquired through experience or education

EXAMPLE The ability to describe the various parts of an information assurance standard.

[SOURCE: ISO/IEC TS 17027:2014, 2.56, modified — an example has been added.]

3.8
review body

entity that reviews the evaluation results provided by an evaluation laboratory for conformance to ISO/IEC 15408

Note 1 to entry: A review body is called an evaluation authority in ISO/IEC 15408 and ISO/IEC 18045.

3.9
review scheme

review system related to specified products, to which the same specified requirements, specific rules and procedures apply

3.10
reviewer

individual assigned to perform review of evaluations in accordance with a given evaluation standard and associated *evaluation methodology* ([3.5](#)) and make decisions based on this

Note 1 to entry: One example of such an individual could be a certifier who uses the evaluation results to perform a certification.

3.11**skill**

ability to perform a task or activity with a specific intended outcome acquired through education, training, experience or other means

EXAMPLE The ability to identify and classify the risks associated with a project.

[SOURCE: ISO/IEC TS 17027:2014, 2.74, modified — an example has been added.]

3.12**validation authority**

entity that validates the testing results provided by a *testing laboratory* (3.3) for conformance to ISO/IEC 19790

Note 1 to entry: A validation authority can modify, add or delete requirements in ISO/IEC 24759.

Note 2 to entry: A validation authority can issue validation certificates for approved security functions or other conformance tested results.

Note 3 to entry: A validation authority can specify the format of the security policy.

Note 4 to entry: A validation authority can use alternative security functions or standards in ISO/IEC 19790:2025, Annexes C to G.

3.13**validation scheme**

validation system related to specified products, to which the same specified requirements, specific rules and procedures apply

3.14**validator**

individual assigned to perform validation activities in accordance with a given conformance testing standard and associated testing methodology

Note 1 to entry: An example of such a conformance testing standard is ISO/IEC 19790 and the testing methodology is specified in ISO/IEC 24759.

4 Concepts

In order to support conformity in the evaluation or conformance-testing of IT security products, competence is one factor in ensuring conformity in the evaluation or conformance-testing of IT security products. Despite the provision of standardized conformance-testing or evaluation methods, a minimum competence in performing the necessary activities is required to support achieving conformity and repeatability of the results. This, in turn, supports the mutual recognition of IT product security assurance certifications and validations.

The ISO/IEC 23532 series address the requirements for the competence of IT security testing and evaluation laboratories, which are related to the competencies of testers and evaluators that are required to be met by a laboratory. These related requirements include ensuring the competence of all personnel who can influence the testing or evaluation laboratory's activities.

Examples of such requirements include the following:

- defining and documenting the competence requirements for each function involved in the testing or evaluation laboratory activities;
- ensuring laboratory personnel have the competence to execute the activities for which they are responsible and understand the significance of and response to deviations found with regard to the testing or evaluation laboratory activities;
- having a documented process for the ongoing monitoring of personnel involved in the testing or evaluation laboratory activities; and

- maintaining records of competence such as for education, training, technical knowledge, skills, experience, authorizations and monitoring for all personnel involved in the testing or evaluation laboratory activities.

NOTE The ISO/IEC 23532 series complement and supplement the procedures and general requirements found in ISO/IEC 17025. The ISO/IEC 23532 series cover a broad range of calibration and testing laboratories and are used in the field of IT product security testing and evaluation.

The requirements for the competence of validation authorities or review bodies are related to competencies of validators or reviewers. These related requirements include ensuring the competence of all personnel who can influence the validation authority's or review body's activities.

Examples of such requirements include the following:

- defining and documenting the competence requirements for each function involved in the validation authority's or review body's activities;
- ensuring the personnel have the competence to execute the activities for which they are responsible and understand the significance of and response to deviations found with regard to the validation authority's or review body's activities;
- having a documented process for the ongoing monitoring of personnel involved in the validation authority's or review body's activities; and
- maintaining records of competence such as for education, training, technical knowledge, skills, experience, authorizations and monitoring for all personnel involved in the validation authority's or review body's activities.

5 Elements of competence

5.1 Competences

In order to competently provide consistent conformance-testing and evaluation results and to support the goal of conformity in the results provided by different personnel and laboratories, it is necessary for conformance-testers and evaluators to have gained the minimum necessary knowledge and skills relevant to the target IT product security assurance standard, and to be able to perform their duties with effectiveness.

This clause defines the minimum elements of competence that are required by conformance-testers, evaluators, validators and reviewers for specific IT product security assurance standards.

Training can be provided in order to increase some elements of competence in personnel. For example, training is often performed to acquire or enhance existing skills or increase knowledge.

Additional elements of competence such as aptitude, initiative, leadership, teamwork and willingness can be specified by laboratories or bodies involved in the application of IT security conformance assessment standards.

5.2 Knowledge

The possession of knowledge by validators, reviewers, testers and evaluators is one of the elements of competence. An appropriate and testable body of knowledge relevant for conformance-testing, evaluation, validation or review shall include:

- a) the relevant IT product security assurance standard;
- b) any associated testing or evaluation methods;
- c) any associated validation or review methods;
- d) policies and procedures of relevant validation authorities, review bodies, and laboratories in which validators, reviewers, testers and evaluators are active;

- e) IT product architecture and design in relevant technology areas; and
- f) IT product life cycles (e.g. manufacture, testing, delivery, operation and end of life).

When considering IT products, a variety of technologies can be pertinent to the scope of work of a laboratory, a validation authority or a review body, and knowledge of these technologies should be considered when defining minimum competency levels. For a particular technology area, the following are important relevant knowledge classes:

- g) the technology used in the design, development and deployment of the products being tested, evaluated, reviewed or validated;
- h) the way in which the products are used or intended to be used;
- i) the typical vulnerabilities and weaknesses which can occur in that technology; and
- j) the domain in which the products are used or intended to be used.

Examples of technology areas include cryptographic module, biometric systems, integrated circuits, operating systems, network devices, databases, smartcards and embedded systems. Technology areas are sometimes defined by the validation authority or review body.

5.3 Skills

5.3.1 General

The skills expected of laboratory, validation authority and review body personnel depend on the technology area and competency level. [5.3.2](#) addresses the skills for testers and evaluators and [5.3.3](#) addresses the skills for validators and reviewers.

5.3.2 Testers and evaluators

Typical skills for laboratory personnel (testers and evaluators) shall include:

- a) understanding the scope and basis of a conformance-testing or an evaluation project;
- b) understanding the boundaries of the implementation under test or the target of evaluation;
- c) ability to select or adapt an appropriate testing or evaluation method;
- d) performing documentation analysis;
- e) understanding the source code, schematics and base components used in specifying and implementing products;
- f) vulnerability analysis;
- g) defining test/evaluation approaches and gathering results with a focus on sufficiency for the intended test/evaluation statement;
- h) ability to build a testing environment to achieve a proper running condition for the implementation under test;
- i) developing and performing functional and non-functional testing;
- j) determining if test conditions are within stated parameters to allow for repeatable testing;
- k) calibrating and using testing tools;
- l) prioritizing the test items and procedures for more efficient, effective and faster testing procedures, while covering all requirements;
- m) interpreting the test results;

- n) ability to write understandable reports detailing the results of their work; and
- o) ability to repeat a test, or replay an archived test, and obtain the same results.

NOTE Vulnerability analysis is applicable depending on scheme.

Integrity, availability and confidentiality of test evidence, test results, and test records, including interpretations and test reports are ensured, therefore the use of adequate storage is expected.

At higher competency levels, communication and project management skills can also be expected.

5.3.3 Validators and reviewers

Typical skills for validation authority and review body personnel (validators and certifiers) shall include:

- a) understanding the scope and basis of a validation or a review project;
- b) understanding the boundaries of the implementation under test or the target of evaluation;
- c) ability to select or adapt appropriate validation or review method;
- d) performing documentation analysis;
- e) understanding the source code, schematics and base components used in specifying and implementing products;
- f) scrutinizing test/evaluation approach and results with a focus on sufficiency for the intended test/evaluation statement;
- g) understanding the test results;
- h) ability to write concise comments on the test or evaluation reports;
- i) ability to write understandable reports detailing the results of their work; and
- j) ability to repeat a review or validation, or replay an archived review or validation, and obtain the same results.

Integrity, availability and confidentiality of test evidence, test results, and test records, validation records and review records including interpretations and test reports are ensured, therefore the use of adequate storage is expected.

At higher competency levels, communication and project management skills can also be expected.

6 Competency levels

6.1 General

Testers, evaluators, validators and reviewers may be assigned a competency level for each specific competence area given in the other parts of the ISO/IEC 19896 series. These levels are described in [6.2](#) and [6.3](#) by a level number and a typical descriptor.

Competency levels for testers and evaluators are described in [6.2.1](#) to [6.2.3](#). Competency levels for validators and reviewers are described in [6.3.1](#) to [6.3.3](#). Level definitions are described in [6.2](#) and [6.3](#) by a level number which may be a subset of competence levels.

6.2 Testers and evaluators

6.2.1 Competency level 1

Testers and evaluators are expected to fulfil the following criteria at competency level 1:

- has knowledge on one of several areas of the validation or evaluation project but cannot demonstrate skills yet;
- provides support for activities required by the conformance-testing or evaluation methods; and
- can perform testing or evaluation work under supervision.

6.2.2 Competency level 2

Testers and evaluators are expected to fulfil the following criteria at competency level 2:

- is competent to work unsupervised in some testing or evaluation areas but can require supervision in a few areas;
- can understand the significance of and response to deviations found with regard to the laboratory's activities; and
- can supervise and provide mentorship in regard to the testing or evaluation work of those at competency level 1.

6.2.3 Competency level 3

Testers and evaluators are expected to fulfil the following criteria at competency level 3:

- is competent in reviewing or consulting on all aspects of the conformance-testing or evaluation according to the defined standards and methods for at least one technology area;
- is competent in communicating with stakeholders including validation authorities or review bodies and vendors and can provide project management for at least part of a validation or evaluation project;
- is competent to work unsupervised in all testing/evaluation methods specified for at least one part of the project;
- can understand the significance of and response to deviations found with regard to the laboratory's activities for at least one technology area; and
- can supervise and provide mentorship in regard to the conformance-testing or evaluation work of those at competency levels 1 and 2.

6.3 Validators and reviewers

6.3.1 Competency level 1

Validators and reviewers are expected to fulfil the following criteria at competency level 1:

- has knowledge on one of several areas of the validation or evaluation project but cannot demonstrate skills yet;
- provides support for activities required by the validation or review methods; and
- can perform validation or review work under supervision.

6.3.2 Competency level 2

Validators and reviewers are expected to fulfil the following criteria at competency level 2:

- is competent to work unsupervised in some validation, review, conformance-testing or evaluation areas but can require supervision in a few areas; and
- can understand the significance of and response to deviations found with regard to the validation authority's or review body's activities.
- can supervise and provide mentorship in regard to the validation or review work of those at competency level 1.

6.3.3 Competency Level 3

Validators and reviewers are expected to fulfil the following criteria at competency level 3:

- is competent in reviewing or consulting on all aspects of the validation, review, conformance-testing or evaluation according to the defined standards and methods for at least one technology area;
- is competent in communicating with stakeholders including laboratories and vendors and can provide project management for at least part of a validation or evaluation project;
- is competent to work unsupervised in all validation/review methods specified for at least part of the project;
- can understand the significance of and response to deviations found with regard to the validation authorities or the review body's activities for at least one technology area; and
- can supervise and provide mentorship in regard to the validation or review work of those at competency levels 1 and 2.

7 Measurement of elements of competence

7.1 Knowledge

The knowledge areas defined in the ISO/IEC 19896 series provide, for each IT product security assurance standard, a body of knowledge that is measurable. Such evidences of knowledge can include professional qualifications gained through third parties or through testing developed and performed by validation authorities, review bodies or the laboratory itself.

7.2 Skills

Various skills are required for validators, reviewers, testers and evaluators of IT security products, which are presented in the other parts of the ISO/IEC 19896 series. These skills should be measured.

The following methods can be used to measure the skills for testers and evaluators:

- performing aspects of the laboratory's proficiency-testing programme as part of the requirements for conformance to the ISO/IEC 23532 series;
- using training records, maintained according to the ISO/IEC 23532 series, in the specification of measures of training effectiveness, which, in turn, can demonstrate the mastery of a skill; and
- integrating feedback from other personnel already deemed as competent in a skill.

The following methods can be used to measure the skills for validators and reviewers:

- performing aspects of the body's proficiency-validating or proficiency-reviewing programme, as part of the requirements for that validation authority or review body;

- using training records, maintained according to the requirements for the validation authority or the review body, in the specification of measures of training effectiveness, which, in turn, can demonstrate the mastery of a skill;
- using professional validations or reviews in regard to particular skills; and
- integrating feedback from other personnel already deemed competent in a skill.

7.3 Recording elements of competence

The ISO/IEC 23532 series requires that records of competence for testers or evaluators are maintained by a laboratory. Records of competence for validators or reviewers are maintained by a validation authority or review body. [Annexes A](#) and [B](#) provide examples of frameworks for recording this information.

Annex A (informative)

Framework for describing competence requirements

A.1 Framework of testers or evaluators

[Table A.1](#) provides a structure that can be used by laboratories to define specific competency requirements using the criteria of knowledge and skills for each competency level. The information in [Table A.1](#) reflects [Clause 7](#) as well as any other requirements defined by ISO/IEC 19896-2, ISO/IEC 19896-3, the laboratory, or the validation or review scheme.

ISO/IEC 19896-2 and ISO/IEC 19896-3 provide the specific competence criteria for testers of ISO/IEC 19790 and evaluators of the ISO/IEC 15408 series that can be used in completing [Table A.1](#).

Table A.1 — Competence requirements record for testers or evaluators

Competency level:	
Knowledge area name	Skill name
Knowledge area description	Skill description

A.2 Framework of validators or reviewers

[Table A.2](#) provides a structure that can be used by a validation authority or review body to define specific competency requirements using the criteria of knowledge and skills. The information in [Table A.2](#) reflects [Clause 7](#) as well as any other requirements defined in ISO/IEC 19896-2, ISO/IEC 19896-3, or by the laboratory, or the validation or review scheme.

ISO/IEC 19896-2 and ISO/IEC 19896-3 provide the specific competence criteria for the validator of ISO/IEC 19790 and the reviewer of the ISO/IEC 15408 series that can be used in completing [Table A.2](#).

Table A.2 — Competence requirements record for validators or reviewers

Competency level:	
Knowledge area name	Skill name
Knowledge area description	Skill description

Annex B (informative)

Example records of experience and competence

[Table B.1](#) provides an example structure for recording the experience gained by a reviewer or evaluator of the ISO/IEC 15408 series or a validator or tester of ISO/IEC 19790.

It is possible that the other parts of the ISO/IEC 19896 series identify more specific examples of records.

Laboratories, validation authorities or review bodies should devise a method of indicating project complexity based on their project portfolio.

EXAMPLE “Simple” and “complex” can refer to the evaluation assurance level for evaluations performed using ISO/IEC 18045, or the security level of a cryptographic module tested using ISO/IEC 24759.

Table B.1 — Example record of experience

Experience record						
Name of individual						
Project ID	Project dates	Technical domain	Project complexity	Hours worked on project	Test/evaluation methods performed	Supervised by /comments (N/A is a valid answer)

[Table B.2](#) provides an example structure for recording the competence gained by a reviewer or evaluator of the ISO/IEC 15408 series or a validator or tester of ISO/IEC 19790.

It is possible that the other parts of the ISO/IEC 19896 series identify other examples of records.

Table B.2 — Example record of competence

Competence record			
Name of individual			
Project ID	Knowledge	Skills	Competency Level

Bibliography

- [1] ISO/IEC 15408 (all parts), *Information security, cybersecurity and privacy protection — Evaluation criteria for IT security*
- [2] ISO/IEC 15443, *Information technology — Security techniques — Security assurance framework*
- [3] ISO/IEC 17024, *Conformity assessment — General requirements for bodies operating certification of persons*
- [4] ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories*
- [5] ISO/IEC/TS 17027:2014, *Conformity assessment — Vocabulary related to competence of persons used for certification of persons*
- [6] ISO/IEC 18045, *Information security, cybersecurity and privacy protection — Methodology for IT security evaluation*
- [7] ISO/IEC 19790:2025, *Information security, cybersecurity and privacy protection — Security requirements for cryptographic modules*
- [8] ISO/IEC 23532 (all parts), *Information security, cybersecurity and privacy protection — Requirements for the competence of IT security testing and evaluation laboratories*
- [9] ISO/IEC 24759, *Information security, cybersecurity and privacy protection — Test requirements for cryptographic modules*

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